BUSINESS RESEARCH METHODS — COMPLETE MASTER STUDY GUIDE

Based on the full “BRM - STUDENT REFERENCE NOTES.pdf” supplied in this chat.

PURPOSE

This guide is a complete study version of the substantive content in the 120-page Student Reference Notes. It includes the smaller points that were condensed in the earlier PDF, including research types, design comparisons, questionnaire details, measurement/scaling, data analysis/presentation, report-writing guidelines, and the image-based ethics slides.

1. RESEARCH

Meaning

- “Re” means once more, again, anew.
- “Search” means try to find something by looking or otherwise seeking carefully and thoroughly.
- Research is an Organized and Systematic way of Finding Answers to Questions.

Key words

- Systematic: there is a definite set of procedures and steps; certain things are done in order to get accurate results.
- Organized: research has a structure or method; it is planned, not spontaneous, focused and limited to a specific scope.
- Finding Answers: finding answers is the end of research. Sometimes the answer is “no”, but it is still an answer.
- Questions: questions are central to research. Research focuses on relevant, useful and important questions; without a question research has no focus, drive or purpose.

2. BUSINESS RESEARCH

- Business research is an organised, systematic and objective process of gathering, recording, analysing and interpreting data that provide information to make accurate business decisions.
- It is also defined as a systematic inquiry whose objective is to provide information that will allow managerial problems to be solved.

Management problem, problem, opportunity and decision making

- Management problem: any problem or opportunity that requires a management decision.
- Problem: a condition that occurs when some aspect of organizational performance is less than desirable.
- Opportunity: any situation that has the potential to provide additional beneficial outcomes.
- Decision making: the process through which managers and leaders identify and resolve problems and capitalize on opportunities.

Who does business research?

- Academics
- Students
- Government and commercial organisation
- Consultants
- Managers

Areas of business research

- Accounting
- Finance
- Marketing
- Human Resources
- Operation Research
- Technology
- Agribusiness Management
- Healthcare Management

Key objective

- The key objective is to provide accurate and timely information to top management so that they can make effective decisions.

Five key questions

- Why should we do research?
- What research should be done?
- How should the research be designed to achieve the research objectives?
- How will it be analysed and interpreted the data?
- What will we do with the research?

3. TYPES OF RESEARCH

1. Basic Research

- Also called pure or fundamental research.
- Mainly conducted to increase the knowledge base.
- Driven by interest and a desire to expand knowledge.
- Tends not to be directly applicable to the real world in a direct way, but enhances understanding of the world.
- Generates new ideas, principles and theories in different fields.
- Sometimes described as not having practical applications.

2. Applied Research

- Mainly related to solving practical problems rather than focusing on knowledge expansion.
- Used to find solutions to problems occurring daily and develop new innovative technologies.
- Main aim is to provide better technologies for humans to enhance standard of living.

3. Quantitative Research

- Generally related with the positivist concept.
- Collects and converts data into numerical form so statistical calculations can be made and conclusions drawn.
- Objectivity is vital.
- Researchers try to avoid their presence, behaviour or attitude affecting results and examine methods/results for bias.
- Aim includes developing mathematical models and theories related to phenomena.
- Mainly used in social sciences.

4. Qualitative Research

- Usually related with the social constructivist concept, emphasizing socially constructed reality.
- Records, analyses and attempts to reveal in-depth meaning and significance of human behaviour and experience, including conflicting beliefs, behaviours and emotions.
- Tries to answer why and how of decision-making rather than what and when.
- Data collection and analysis are logical but more flexible than quantitative research.
- Data are collected in textual form based on observation and communication with participants.

5. Descriptive Research

- Used to describe characteristics of an observable fact being studied.
- Structured so it cannot be changed frequently; therefore rigid in nature.
- Cannot identify cause-and-effect relationship between variables.
- Answers who, when, where, what and how.
- Describes what exists and may reveal new facts and meanings.
- Purpose: observe, describe and document.

6. Exploratory Research

- Carried out for a problem that has not been clearly defined.
- Gathers initial information to help define problems and recommend/formulate hypotheses.
- Helps settle the best research design, data collection method and selection of subjects.
- May rely on secondary research/literature and qualitative approaches: informal discussions with consumers, employees, management or competitors; in-depth interviews; focus groups; projective methods; case studies; pilot studies.
- Conducted when researchers lack a clear idea of the problem.
- Results are not generally useful for decision-making, but can provide major insight.

7. Historical Research

- Examines past events or combinations of events to arrive at an account of what happened in the past.
- Used to discover the unknown; answer questions; recognize relationship of past to present; record and assess activities of individuals, agencies or institutions; and understand culture.
- Can show patterns that occurred in the past and over time, helping us see where we came from and solutions used in the past.
- What we do today is often rooted in the past.
- Involves collecting and reading research material and writing the document from collected data.

8. Experimental Research

- Commonly used in sociology, psychology, physics, chemistry, biology, medicine, etc.
- Uses manipulation and controlled testing to understand fundamental processes.
- Usually one or more variables are manipulated to establish their effect on a dependent variable.
- Mainly used when there is time priority in a causal relationship (cause precedes effect), uniformity in a causal relationship (a cause will always lead to the same effect), or a great magnitude of correlation.
- Important to society because it helps improve daily life.

9. Causal Research

- Also called explanatory research.
- Conducted to identify extent and nature of cause-and-effect relationships.
- Can assess impacts of specific changes on existing norms and processes.
- Focuses on analysing a situation/specific problem to explain patterns of relationships between variables.

Causal vs Exploratory Research

- Causal: decision situation clearly defined; key statement is research hypotheses; later stages of decision making; highly structured.
- Exploratory: decision situation highly ambiguous; key statement is research question; early stage of decision making; unstructured.
- Causal examples: “Will consumers buy more products in a blue package?” and “Which of two advertising campaigns will be more effective?”
- Exploratory examples: “Our sales are declining for no apparent reason” and “What kinds of new products are fast-food consumers interested in?”

10. Theoretical Research

- Enables exploration and discussion of a research object based on abstract theoretical structures and philosophical concepts.
- Research object cannot be directly observed.
- Aim is to define and outline conceptual models, explanations and structures using research literature.

11. Empirical Research

- Research where conclusions are strictly drawn from concrete empirical evidence and therefore verifiable evidence.
- Evidence can be gathered using quantitative and qualitative market research methods.

12. Cross-sectional Research

- Uses different groups of people who differ in the variable of interest but share other characteristics such as socioeconomic status, educational background and ethnicity.
- Often used in developmental psychology.
- Also used in social science and education.
- Participants are selected based on variables of interest.

13. Time Series Research

- Time series analysis is a statistical technique dealing with time series data or trend analysis.
- Time series data are data in a series of time periods or intervals.
- Time series data are a set of observations on values a variable takes at different times.

Qualitative vs Quantitative Research

- Focus: Qualitative explores ideas/formulates hypotheses or theories; quantitative tests hypotheses or theories.
- Analysis: Qualitative summarises, categorises and interprets; quantitative uses mathematical and statistical analysis.
- Expressed in: Qualitative words; quantitative numbers, graphs, tables and fewer words.
- Sample: Qualitative few respondents; quantitative many respondents.
- Questions: Qualitative open-ended; quantitative close-ended or multiple choice.
- Characterized by: Qualitative understanding, context, complexity, subjectivity; quantitative testing, measurement, objectivity, replicability.

4. BUSINESS RESEARCH PROCESS

Main flow

Defining the Problem → Developing an Approach to the Problem → Research Design → Data Collection → Data Analysis and Interpretation → Preparing and Presenting the Report → Management Decision.

Detailed process

1. Selection of a research problem or researchable area.

2. Literature review.

3. Definition and specification of the research problem more clearly.

4. Formulation of research hypothesis or at least research objectives.

5. Preparing research design, including identification of sources of data and sample design.

6. Creation and construction of data-collection instruments such as questionnaires and scales.

7. Pilot testing of instruments and possible revision.

8. Formal acquisition of data/information through survey, observation, interview, etc.

9. Processing and analysis of data.

10. Interpretation of data and formal write-up/reporting.

11. Management decision.

5. RESEARCH DESIGN

Definition

- A research design is a framework or blueprint for conducting the marketing research project.
- It details procedures necessary for obtaining information needed to structure or solve marketing research problems.

Components of research design

- Define information needed.
- Design exploratory, descriptive and/or causal phases.
- Specify measurement and scaling procedures.
- Construct and pretest questionnaire/interviewing form or appropriate data-collection form.
- Specify sampling process and sample size.
- Develop plan of data analysis.

Study/design components listed in notes

1. Statement of Problem

2. Need for the Study

3. Review of Previous Studies

4. Objectives of Study

5. Formulation of Hypothesis

6. Operational Definitions

7. Scope of the Study

8. Sources of Data

9. Sampling Design

10. Tools & Techniques

Classification of marketing research designs

- Research Design → Exploratory Research Design + Conclusive Research Design.
- Conclusive includes Descriptive Research and Causal Research.
- Descriptive includes Cross-Sectional Design and Longitudinal Design.
- Cross-Sectional includes Single Cross-Sectional Design and Multiple Cross-Sectional Design.

Exploratory vs Conclusive

- Exploratory objective: provide insights and understanding.
- Exploratory information needed: loosely defined.
- Exploratory process: flexible and unstructured.
- Exploratory sample: small and non-representative.
- Exploratory primary-data analysis: qualitative.
- Exploratory findings: tentative; generally followed by further exploratory or conclusive research.
- Conclusive objective: test specific hypotheses and examine relationships.
- Conclusive information needed: clearly defined.
- Conclusive process: formal and structured.
- Conclusive sample: large and representative.
- Conclusive analysis: quantitative.
- Conclusive outcome: findings used as input into decision making.

Comparison of basic research designs

Exploratory

- Objective: discovery of ideas and insights.
- Characteristics: flexible, versatile; often front end of total research design.

- Methods: expert surveys, pilot surveys, secondary data analysed qualitatively, qualitative research.

Descriptive

- Objective: describe market characteristics or functions.
- Characteristics: prior formulation of specific hypotheses; preplanned and structured design.
- Methods: secondary data analysed quantitatively, surveys, panels, observation and other data.

Causal

- Objective: determine cause-and-effect relationships.
- Characteristics: manipulation of one or more independent variables; control of other mediating variables.
- Method: experiments.

Uses of exploratory research

- Formulate/define a problem more precisely.
- Identify alternative courses of action.
- Develop hypotheses.
- Isolate key variables and relationships for further examination.
- Gain insights for developing an approach to the problem.
- Establish priorities for further research.

Methods of exploratory research

- Survey of experts
- Pilot surveys
- Secondary data analysed qualitatively
- Qualitative research

Uses of descriptive research

- Describe characteristics of relevant groups: consumers, salespeople, organizations or market areas.
- Estimate percentage of units in a specified population exhibiting a behavior.
- Determine perceptions of product characteristics.
- Determine degree to which marketing variables are associated.
- Make specific predictions.

Methods of descriptive research

- Secondary data analysed quantitatively.
- Surveys.
- Panels.
- Observational and other data.

6. DATA COLLECTION AND QUESTIONNAIRE DESIGN

Data

- Data means information about a particular phenomenon.
- Two types: Primary Data and Secondary Data.

Primary Data

- Original data collected specially for the purpose in mind.
- Collected for the first time by an authorized organization, investigator or enumerator, alone or with institution/expert.
- Important because it is unvarnished information about results of an experiment or observation; notes compare it to eyewitness testimony.

Advantages of Primary Data

- Degree of accuracy is quite high.
- Does not require extra caution.
- Depicts data in great detail.
- Primary source frequently includes definitions of terms and units used.
- For some investigations, secondary data are not available.

Disadvantages of Primary Data

- Requires a lot of time.
- Requires a lot of finance.
- Sometimes not possible to collect.
- Requires a lot of labour.

Secondary Data

- Notes describe it as data collected for another purpose and also as data already collected and readily available from other sources.
- Such data are cheaper and more quickly obtainable than primary data and may be available when primary data cannot be obtained.

Advantages of Secondary Data

- Economical; saves effort and expense.
- Helps make primary data collection more specific.
- Helps identify gaps/deficiencies and additional information needed.
- Improves understanding of the problem.
- Provides a basis for comparison for researcher-collected data.

Disadvantages of Secondary Data

- May seldom fit the marketing-research framework.
- Unit of data may differ from what is required, e.g. disposable income desired but gross income available.
- Class boundaries may differ even when units are the same.
- Earlier data may therefore be of no use.
- Accuracy may not be known.
- Data may be outdated.

Primary vs Secondary Data

- Primary: more funds; investing agency collects; longer time; more reliable/suitable because investigator collects; elaborate organization.
- Secondary: comparatively less funds; some other agency collected it for own use; less time; may be less reliable/suitable because collection may not fit purpose; no organization setup needed.

Primary data collection methods

1. Observation Method

2. Survey Method

3. Questionnaire

4. Schedule

Observation Method

- Human or mechanical observation of what people actually do or what events take place during buying/consumption; information is collected by observing process at work.

Examples

- Service station: pose as customer and observe.
- Dunlop Pillow Cushions display: observe how many pass by, stop to look and decide to buy.
- Supermarket: use hidden cameras to determine best shelf location.
- Salesmen: investigator may use concealed tape recorder to determine sales arrangement and enthusiasm.

Types

- Participant
- Non-participant

Advantages

- Does not rely on willingness/ability of respondents to report accurately.
- Interview bias is eliminated or reduced.
- Data are more objective and generally more accurate.

Disadvantages

- Cannot easily observe attitudes, motivations, state of mind, buying motives and images.
- Takes time to wait for a particular action.
- Personal/intimate activities are more easily discussed in questionnaires than observed.
- Usually more expensive than other survey data.
- Observer may wait unproductively between events, increasing cost.

Survey Method

- Technique of gathering data by asking questions to people thought to have desired information.
- Formal list of questionnaire is prepared.
- Generally non-disguised approach.
- Respondents asked about demographic information, interests and opinions.

Advantages

- Broader range of information than direct observation/experimentation.
- Effective for socio-economic characteristics, attitudes, opinions, motives and planning product features, advertising media, sales promotion, channels of distribution and other marketing variables.
- Usually faster and cheaper than observation.
- Questions simple to administer.
- Data reliable.

- Variability of results reduced.
- Relatively simple to analyse, quote and interrelate.

Disadvantages

- Unwillingness to provide information; interviewer may need salesmanship.
- Interviewer may assure secrecy or offer presents.
- Inability due to lack of knowledge, lapse of memory, inability to identify motives/reasons.
- Human biases such as ego.
- Semantic difficulties: question may not mean exactly the same to every respondent.
- Different wording can produce different results.

Survey techniques

- Telephonic Interview
- Personal Interview
- Mail Interview
- Electronic Interview

Questionnaire

- Formalized set of questions for obtaining information from respondents.
- Must translate information needed into specific questions respondents can and will answer.
- Must motivate and encourage involvement, cooperation and completion.
- Should not be too lengthy.
- Use simple English; questions should not be difficult to answer.
- Requires sensible language, editing, assessment and redrafting.

Questionnaire Design Process

1. Specify information needed.

2. Design questions to overcome inability and unwillingness to answer.

3. Determine content of individual questions.

4. Decide question structure.

5. Determine question wording.

6. Arrange questions in proper order.

7. Reproduce questionnaire.

8. Specify interviewing method.

9. Identify form and layout.

10. Eliminate bugs by pre-testing.

Detailed questionnaire design

1. State information required

- Depends on nature of problem, purpose of study and hypothesis.
- Concentrate on target audience.

2. State interviewing technique

- Telephone, mail, personal or electronic interview.

3. Decide content of individual questions

- Is the question significant and does it contribute to the study objective?
- Is one question enough or are several needed?
- Several questions may be needed for cross-checking, ambiguous answers, or hesitant respondents.

4. Overcome inability/unwillingness

- Inability: not fully informed, may not remember, may be unable to express/articulate.
- Unwillingness: sensitive information may cause embarrassment/harm to image; respondent may not know genuine purpose; question may appear irrelevant.
- Solutions: sensitive topics at end; preface questions with a statement; use third-person technique.

5. Structure

- Structured questions specify response alternatives and format; include multiple-choice, dichotomous questions and scales.
- Unstructured/open-ended questions provide no alternatives; respondents answer freely.

6. Language/phrasing

- Use six Ws: who, what, when, where, why and way.
- Use ordinary words matching vocabulary level.
- Avoid ambiguous words: usually, normally, frequently, often, regularly, occasionally, sometimes.
- Avoid leading questions.
- Do not require generalizations or computed estimates.
- Use positive and negative statements.

7. Order

- Opening questions interesting, simple and non-threatening.
- Qualifying questions should serve as opening questions.
- Basic information first, then classification, finally identification information.
- Difficult/sensitive/complex questions late.
- General before specific.
- Logical order.
- Branching questions carefully designed to cover all contingencies.

8. Form and layout

- Divide questionnaire into several parts.
- Number questions, especially with branching.
- Preferably pre-code.
- Number questionnaires serially.

9. Reproduce

- Good-quality paper and professional appearance.
- Prefer booklet to clipped/stapled sheets.
- Each question on one page or double-page spread.
- Vertical response columns for individual questions.
- Grids for related questions with same response categories.
- Avoid crowding to make it look shorter.
- Put directions close to relevant questions.

10. Pre-test

- Always pretest.
- Test content, wording, sequence, form/layout, difficulty and instructions.
- Respondents should be similar to actual survey respondents.
- Begin with personal interviews.
- Also pretest by mail/telephone if those methods will be used.
- Use a variety of interviewers.
- Initial sample about 15–30 respondents.
- Use protocol analysis and debriefing.
- After each significant revision, conduct another pretest with a different sample.
- Code and analyse pretest responses.

Questionnaire vs Schedule

- In practical life they may have the same meaning, but theoretically they are distinguished.
- Questionnaire: respondent/informant fills the form; schedule: investigator/enumerator fills the form after face-to-face questioning.
- Questionnaire: complex questions can create non-response; schedule: notes state no chance of non-response.
- Questionnaire: list/series of questions; schedule: notes also describe schedule as scheduled events/plans for specific days/dates.
- Questionnaire is relatively cheap/economical; schedule is relatively more expensive.
- Questionnaire: identity of person replying is not always clear; schedule: respondent identity is known.

7. MEASUREMENT AND SCALING

Measurement

- Assigning numbers or other symbols to characteristics of objects according to pre-specified rules.
- One-to-one correspondence between numbers and characteristics measured.
- Rules must be standardized and applied uniformly.
- Rules must not change over objects or time.

Scaling

- Creating a continuum upon which measured objects are located.
- Example: attitude scale 1–100; 1 = Extremely Unfavorable, 100 = Extremely Favorable.
- Measurement is assigning a number from 1 to 100; scaling is placing respondents on the continuum regarding attitude.

Continuum

- A continuous range from one extreme of a characteristic to another; example: Extremely Unfavorable → Extremely Favorable.

Measurement vs Scaling

- Measurement = assignment of numbers/symbols according to rules.
- Scaling = creating a continuum and locating objects/respondents on it.

8. PRIMARY SCALES OF MEASUREMENT — NOIR

Nominal

- Numbers are labels/tags for identifying/classifying objects.
- Strict one-to-one correspondence for identification.
- Numbers do not reflect amount of characteristic.
- Only counting is permissible.
- Frequency-based statistics such as percentages and mode.

Ordinal

- Ranking scale showing relative extent of a characteristic.
- Shows more/less, not how much more/less.
- Any series of numbers preserving order can be used.
- In addition to counting, centile statistics such as percentile, quartile and median.

Interval

- Equal numerical distances represent equal values in the characteristic.
- Permits comparison of differences.
- Zero point is not fixed; zero and units are arbitrary.
- Positive linear transformation y = a + bx preserves properties.
- Ratios are not meaningful.
- Can use nominal/ordinal statistics plus arithmetic mean, standard deviation and other common marketing-research statistics.

Ratio

- Has properties of nominal, ordinal and interval scales.
- Absolute zero point.
- Ratios are meaningful.
- Only proportionate transformation y = bx where b is positive is allowed.
- All statistical techniques can be applied.

Scale summary from notes

- Nominal: identifies/classifies; examples Social Security numbers, football player numbers, brand numbers, store types; descriptive: percentages/mode; inferential: chi-square/binomial test.
- Ordinal: relative positions not magnitude of differences; examples quality rankings, tournament team rankings, preference rankings, market position, social class; descriptive: percentile/median; inferential: rank-order correlation/Friedman ANOVA.
- Interval: differences between objects; examples Fahrenheit temperature, attitudes/opinions/index; descriptive: range/mean/standard deviation; inferential: product-moment.
- Ratio: fixed zero and comparable ratios; examples length/weight, age/sales/income/costs; descriptive: geometric mean/harmonic mean; inferential: coefficient of variation.

9. SCALING TECHNIQUES

Classification

- Comparative scales: Paired Comparison, Rank Order, Constant Sum.
- Noncomparative scales: Continuous Rating, Itemized Rating.
- Itemized rating includes Likert, Semantic Differential and Stapel.

Comparative scales

- Direct comparison of stimulus objects.
- Data interpreted in relative terms and have ordinal/rank-order properties.

Advantages

- Small differences between stimulus objects can be detected.
- Same known reference points for all respondents.
- Easily understood and applied.
- Fewer theoretical assumptions.
- Tend to reduce halo/carryover effects from one judgment to another.

Disadvantages

- Ordinal nature of data.
- Cannot generalize beyond stimulus objects scaled.

Paired Comparison Scaling

- Respondent is presented with two objects and selects one according to a criterion.
- Data are ordinal.
- Most widely-used comparative scaling technique in notes.
- With n brands, comparisons = n(n–1)/2.
- Under transitivity, paired comparison data can be converted to rank order.

Example: 4 brands → 4×3/2 = 6 comparisons: A-B, A-C, A-D, B-C, B-D, C-D.

Rank Order Scaling

- Several objects presented simultaneously; respondent orders/ranks them according to a criterion.
- Brand ranked #1 may still be disliked in absolute terms.
- Produces ordinal data.
- Only n–1 scaling decisions are needed.

Constant Sum Scaling

- Allocate a constant sum such as 100 points to product attributes according to importance.
- Unimportant attribute gets zero.
- Attribute twice as important receives twice as many points.
- Total = 100, hence constant sum.

Noncomparative Scaling

- Respondents evaluate one object at a time; also called monadic scales.
- Consists of continuous and itemized rating scales.

Continuous Rating Scale

- Respondent marks the appropriate position on a line running from one extreme to another.
- Form may vary.
- Example: rate D-Mart from “Probably the worst” to “Probably the best”, with versions using 0–100 and “Very bad / Neither good nor bad / Very good”.

Itemized Rating Scales

- Number or brief description associated with each category.
- Categories ordered by scale position.
- Respondent selects category that best describes object.
- Common scales: Likert, semantic differential, Stapel.

Likert Scale

- Respondents indicate degree of agreement/disagreement with a series of statements.
- Notes show 1–5: Strongly Disagree, Disagree, Neither Agree nor Disagree, Agree, Strongly Agree.
- Analysis may be item-by-item (profile analysis) or total/summated score.
- For negative statements, reverse the scale when calculating total score.

Semantic Differential Scale

- Seven-point rating scale with bipolar labels at endpoints.
- Examples: Powerful—Weak; Unreliable—Reliable; Modern—Old-fashioned.
- Negative adjective/phrase may be placed left or right to control tendency to mark a side without reading labels.
- Individual items may be scored –3 to +3 or 1 to 7.

Stapel Scale

- Unipolar rating scale with ten categories numbered –5 to +5, without neutral zero.
- Usually presented vertically.
- Data can be analysed in same way as semantic differential data.

Basic noncomparative scale comparison

- Continuous rating: mark on continuous line; example reaction to TV commercials; advantage easy to construct; disadvantage scoring cumbersome unless computerized.
- Likert: degree of agreement 1–5; attitude measurement; easy to construct/administer/understand; more time-consuming.
- Semantic Differential: 7-point bipolar labels; brand/product/company images; versatile; controversy whether data are interval.
- Stapel: unipolar –5 to +5 without neutral zero; attitudes/images; easy to construct and administer by telephone; confusing/difficult to apply.

10. DATA ANALYSIS, INTERPRETATION AND REPORT WRITING

Why analyze data?

- Obtain usable and useful information.
- Describe and summarize data.
- Identify relationships between variables.
- Compare variables.

- Identify differences between variables.
- Forecast outcomes.

Data analysis vs interpretation

- Data analysis: processing and analysing data to derive meaningful insights.
- Data interpretation: making sense of insights and drawing conclusions.

Data analysis techniques

- Descriptive Statistics: summarise/describe data using mean, median, standard deviation.
- Inferential Statistics: make inferences about population based on sample; includes hypothesis testing, confidence intervals and regression analysis.
- Content Analysis: analyse text, images or videos to identify patterns and themes.
- Data Mining: use statistical and machine-learning techniques to analyse large datasets and identify patterns.

Data interpretation techniques

- Data Visualization: present data visually using charts, graphs or tables.
- Storytelling: present insights in narrative form to make them relatable and memorable.
- Comparative Analysis: compare findings with existing literature or benchmarks.

Data presentation techniques

- Tables and Graphs: effective for numerical data.
- Infographics: present complex data visually and easily.
- Data Storytelling: present data in narrative form to communicate findings.

11. HYPOTHESIS TESTING

- Statistical method used to make decisions about a population based on sample data.
- Helps test assumptions and draw conclusions from data.

Importance

- Testing assumptions: avoids inaccurate/misleading conclusions.
- Making informed decisions: supports accurate, reliable and valid findings.
- Replicability: helps replicate findings and assess validity/reliability.

Basic concepts

- Null Hypothesis (H0): assumes no significant difference between sample data and population data.
- Alternative Hypothesis (H1): assumes significant difference between sample data and population data.
- Significance Level: measure of strength of evidence against H0; usually 0.05 or 0.01.

Procedure

1. Define H0 and H1.

2. Choose significance level, usually 0.05 or 0.01.

3. Calculate test statistic; choice depends on research question/objectives; notes mention t-tests, ANOVA and chi-square.

4. Calculate p-value.

5. Interpret: if p-value < significance level, reject H0 in favour of H1.

12. RESEARCH REPORT WRITING

Importance

- Written report is documentary/historical evidence after project completion and management decision.
- Management decisions are guided by report and presentation.
- Many decision makers only read report and listen to presentation; they evaluate project quality through them.
- Future research/supplier decisions may be influenced by usefulness of report/presentation.

Report preparation and presentation process

Problem Definition, Approach, Research Design and Fieldwork → Data Analysis → Report Preparation → Interpretations, Conclusions and Recommendations → Oral Presentation → Reading of Report by Client → Research Follow-Up.

- Report formats may vary with researcher or marketing research firm.

Report format

I. Title page

II. Letter of transmittal

III. Letter of authorization

IV. Table of contents

V. List of tables

VI. List of graphs

VII. List of appendices

VIII. List of exhibits

IX. Executive summary: a) Major findings b) Conclusions c) Recommendations

X. Problem definition: a) Background to problem b) Statement of problem

XI. Approach to problem

XII. Research design: a) Type of research design b) Information needs c) Secondary data collection d) Primary data collection e) Scaling techniques f) Questionnaire development and pretesting g) Sampling techniques h) Fieldwork

XIII. Data analysis: a) Methodology b) Plan of data analysis

XIV. Results

XV. Limitations and caveats

XVI. Conclusions and recommendations

XVII. Exhibits: a) Questionnaires and forms b) Statistical output c) Lists

Good report writing

- Readers: write for specific readers, especially managers who use results. Avoid technical jargon unless appropriate; if unavoidable, define it in glossary/appendix.
- Easy to follow: logical structure, clear writing, headings and subheadings.
- Presentable and professional: quality paper, typing and binding for hard copies; skilful graphic design online; use type size and white space carefully; too much variation can confuse.
- Objective: accurately present design, results and conclusions without slanting findings to management expectations; researcher should defend results objectively.

- Reinforce text with tables and graphs: use tables, graphs, pictures, maps and visual devices for clarity and impact.
- Conciseness: omit unnecessary material; avoid lengthy common procedures; brevity must not sacrifice completeness.

Guidelines for Tables

- Title and number: every table should have a number and title.
- Arrangement of data items: emphasize the most significant aspect.
- Basis of measurement: state basis/unit clearly.
- Leaders, rulings, spaces: leaders/dots/hyphens guide the eye; white space can separate data; skipping lines can help; horizontal rules often follow headings.
- Headings, stubs and footnotes: headings are over vertical columns; stubs are in left-hand column; footnotes explain information not incorporated in table.
- Sources: cite source when table data are secondary.

13. RESEARCH & ETHICS

The final ethics material appears as rendered slides in the source PDF; the following is the slide content.

Ethics

- Derived from Greek word Ethos meaning custom, convention and spirit of community.
- Refers to moral principles governing a person’s behaviour.
- Branch of philosophy dealing with morality.
- Concerned with distinguishing good/evil, right/wrong human action, acceptable/unacceptable behaviour, and virtuous/non-virtuous characteristics.

Research Ethics

- Set of principles governing the way research should be designed, managed and conducted.
- Provides guidelines for responsible conduct of research.
- Educates and monitors scientists conducting research to ensure a high ethical standard.

Why is Research Ethics Important?

- Protect vulnerable groups and other study participants.
- Safeguard participants from exploitation.
- Ensure sufficient knowledge is shared by all concerned.
- Establish risk-benefit ratio for study subjects.
- Ensure respect, privacy, disclosure and fair treatment of subjects/participants.
- Build capability of subjects to accept or reject participation.
- Promote research aims such as knowledge, truth and avoidance of error.
- Promote values for collaborative work.
- Build public support for research.

General ethical principles with human subjects

Principle of Autonomy

- Right of an individual to determine what activities to participate in or not.
- Individual should understand what is being asked and make a reasoned judgment about effect of participation.
- Choice should be free of coercive influence.

Principle of Beneficence

- Obligation of investigator to maximize benefits for participants/society while minimizing risk of harm.

Principle of Justice

- Requires equitable selection of participants.
- Subjects involved in research are beneficiaries of research findings.

Ethical Issues

Voluntary participation

- Subjects should be allowed free will in deciding whether to participate.
- Participants have right to withdraw at any stage.

Informed consent

- Requires open and honest communication between researcher and participant.
- Researcher should provide sufficient information about implications of participation.
- Acceptance should be free of coercion.
- May be written or verbal.

Research language

- Offensive, discriminatory or otherwise unacceptable language should be avoided in formulation of research-instrument items.

Honesty

- Truthfulness/accuracy of researchers' actions.
- Ensure integrity in entire research process.
- Commitment to intellectual honesty and personal responsibility.
- Examples of misconduct: fabrication, falsification, plagiarism.

Confidentiality

- Protect information obtained from participants.
- Conceal identity of subjects, e.g. by study codes.

Respect for participants

- Concerned with wellbeing of subjects.
- Examples: assuring privacy/confidentiality, sharing results, providing emerging information.

Social responsibility

- Promote social good.
- Prevent or mitigate social harms through research, public education and advocacy.

Objectivity

- State/quality of being true and without individual biases, interpretations, feelings and imaginations.
- Reduce/eliminate biases, prejudices or subjective evaluations by relying on verifiable data.

14. LAST-DAY MASTER MEMORY SHEET

Research = Organized + Systematic + Answers Questions.

Business Research = Objective data process → accurate decisions.

Problem = performance less than desirable.

Opportunity = potential beneficial outcome.

Basic = knowledge/theory.

Applied = practical problem solving.

Qualitative = why/how + words + flexible.

Quantitative = numbers + statistics + objective.

Descriptive = describe what exists; no cause-effect.

Exploratory = unclear problem + initial insight.

Causal = cause-effect.

Historical = past.

Experimental = manipulation + control.

Theoretical = abstract models/concepts.

Empirical = verifiable evidence.

Cross-sectional = groups differing in variable of interest.

Time series = observations across time.

Research Design = blueprint.

Primary = first-hand/original.

Secondary = already collected/available.

Observation = watch behaviour.

Survey = ask questions.

Questionnaire = respondent fills.

Schedule = enumerator/investigator fills.

Measurement = assign numbers.

Scaling = locate on continuum.

Continuum = range from one extreme to another.

NOIR = Nominal → Ordinal → Interval → Ratio.

Paired = two at a time; n(n–1)/2.

Rank Order = rank several; n–1 decisions.

Constant Sum = distribute 100.

Continuous = mark on line.

Likert = agreement/disagreement.

Semantic Differential = 7-point bipolar adjectives.

Stapel = –5 to +5, no neutral zero.

Descriptive Statistics = summarize.

Inferential = infer population from sample.

Content Analysis = text/images/videos → patterns/themes.

Data Mining = large datasets → patterns.

Visualization = charts/graphs/tables.

H0 = no significant difference.

H1 = significant difference.

Alpha = usually .05 or .01.

p < alpha = reject H0.

Report = historical record + decision support.

Good report = reader-focused + clear + professional + objective + visual + concise.

Ethics = moral principles in research.

Autonomy = free choice.

Beneficence = maximize benefit/minimize harm.

Justice = equitable selection.

Consent = informed + free of coercion.

Confidentiality = protect participant information.

Honesty = no fabrication/falsification/plagiarism.

Objectivity = reduce bias using verifiable data.

15. WHAT WAS EASY TO MISS IN THE EARLIER VERSION

- Re + Search and detailed meanings of systematic, organized, finding answers and questions.
- Management problem/opportunity/decision-making definitions.
- Who conducts business research and all eight areas.
- Five key business-research questions.
- Historical, experimental, theoretical, empirical, cross-sectional and time-series research.
- Full qualitative-vs-quantitative comparison.
- Causal-vs-exploratory comparison and examples.
- Full research-design classification and detailed components.
- Exploratory/descriptive uses and methods.
- All primary/secondary advantages and disadvantages.
- Observation examples, types, advantages and disadvantages.
- All four survey methods and detailed limitations.
- Full questionnaire wording, ordering, layout, reproduction and pretesting guidance.
- Pretest details: similar respondents, personal interviews, other modes, variety of interviewers, 15–30 initial respondents, protocol analysis, debriefing, repeat pretests and coding/analysis.
- Measurement rules and scaling/continuum distinction.
- NOIR properties, transformations and statistics.
- Comparative-scale advantages/disadvantages and all three techniques.
- Continuous rating and all itemized rating details.
- Likert profile/summated scoring and reverse scoring.
- Semantic differential label-order rationale and scoring ranges.
- Stapel no-neutral-zero detail.
- Content analysis, data mining, visualization, storytelling and comparative analysis.
- Full report format, good-report characteristics and table guidelines.
- All image-based ethics slides.

SOURCE COVERAGE NOTE

The supplied PDF includes title/transition slides and a final quote slide. Those are not substantive study topics, but all identifiable substantive content has been incorporated. The Research & Ethics section was transcribed from the rendered slide images because the indexed text layer did not contain the slide text.